Project1

Pandas is a powerful Python library used for data manipulation and analysis. It builds on top of NumPy and provides easy-to-use data structures for working with structured data like tables.

Importing files using pandas is one of the most common tasks in data analysis. Pandas provides simple functions to read different file formats into a DataFrame.

Here we are importing a CSV file into Pandas and Pandas returns a Data Frame object.

```
import pandas as pd
df=pd.read_csv("railway_gauges 1.csv")
print(df.head())
```

Figure 1:Loading CSV file into Pandas

Using Pandas Data Frame object we can see **head()**, **tail()** functions to display top 5, last 5 rows from imported dataset.

```
op/Projects/Project1/app.py
      Year  Broad Gauge  Metre Gauge  Narrow Gauge  Total
0  1964-65         3521         2891           464   6876
1  1965-66         3603         2913           470   6986
2  1966-67         3615         2911           481   7007
3  1967-68         3601         2920           506   7027
4  1968-69         3603         2921           508   7032
PS C:\Users\Windows\OneDrive\Desktop\Projects\Project1>
```

Figure 2: Displaying top 5 rows

Some of the Pandas functions are:

- **read_csv()**-Loads CSV file into a DataFrame
- **iloc[]** - used for selecting data by integer position
- **idxmax()**-Returns index of maximum value

```
print(df.iloc[[df["Total"].idxmax()]])
"""----------------------------------------------------
```

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	Year	Broad Gauge	Metre Gauge	Narrow Gauge	Total
48	2012-13	6577	379	216	7172

Figure 3: Displaying details of highest Total Year

I used pandas functions such as read_csv() to load the dataset, head() to preview data, fillna() to handle missing values, and to_numeric() for type conversion. I also used sum() for aggregation and loc[] for filtering. Attributes like columns and dtypes were used to understand the dataset structure. Additionally, I performed column selection, filtering, and feature creation for analysis.

Matplotlib is a popular Python library used for data visualization. It allows you to create charts and graphs to better understand and present data.

It works well with libraries like NumPy and pandas.

Some of the Matplotlib functions are:

- plt.plot() - is used to create a line graph.
- plt.bar() - creates a bar chart.
- plt.show() - Displays the plot.

DATA VISUALIZATION

Matplotlib is used to visualize data in graphical form, helping to analyze trends, compare values, and understand large datasets easily. It is widely used in data science, machine learning, and scientific computing for creating charts like line graphs, bar graphs, and pie charts.

This code is used to visualize the relationship between Year and Number of railway tracks using a bar graph with Matplotlib.

```
df=df.drop("Total",axis=1)
ax=df.plot(x="Year",kind="bar")
plt.xticks(rotation=70)
plt.xlabel("YEAR")
plt.ylabel("TOTAL")
plt.title("Gauges: Number of railway tracks installed per year")
plt.savefig("rail_gauges.png")
plt.show()
```

Figure 3: Bar Graph Code

The resulting graph shows how the number of railway tracks changes over different years, helping to identify trends such as increase or decrease over time.

RESULTS

The project successfully analyzes railway gauge data using pandas and Matplotlib, and concludes that Broad Gauge dominates the modern railway system with overall increasing track growth.

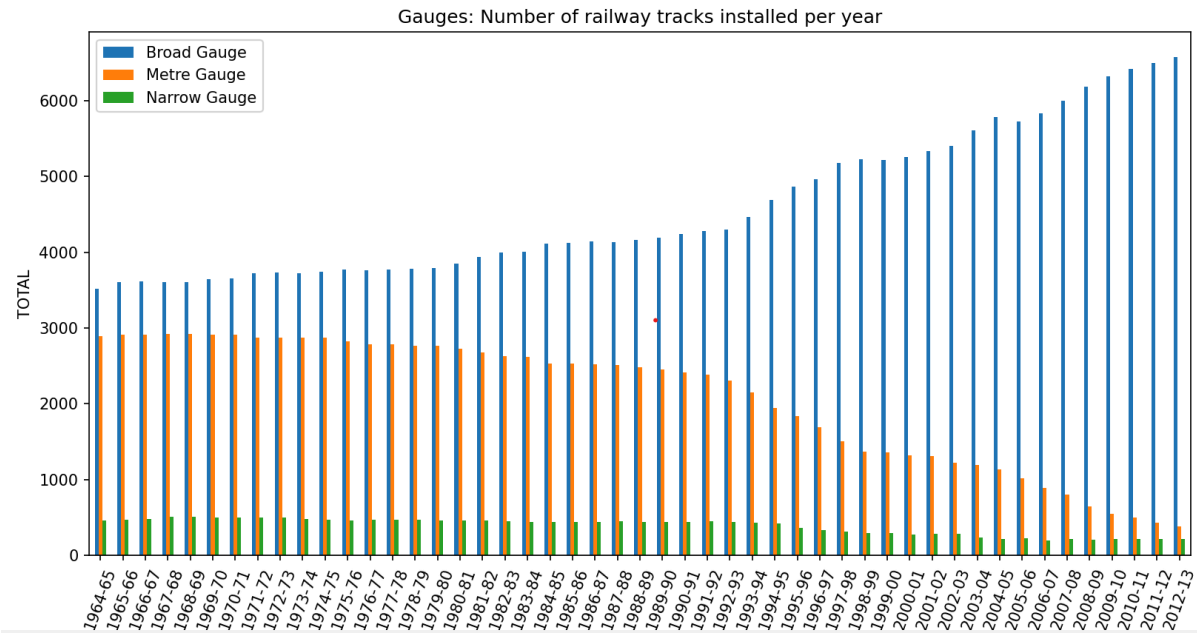

Figure 4: Number of Railway tracks installed per year